Solar Equipment Maintenance Assistant

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Research Motivation & Objectives

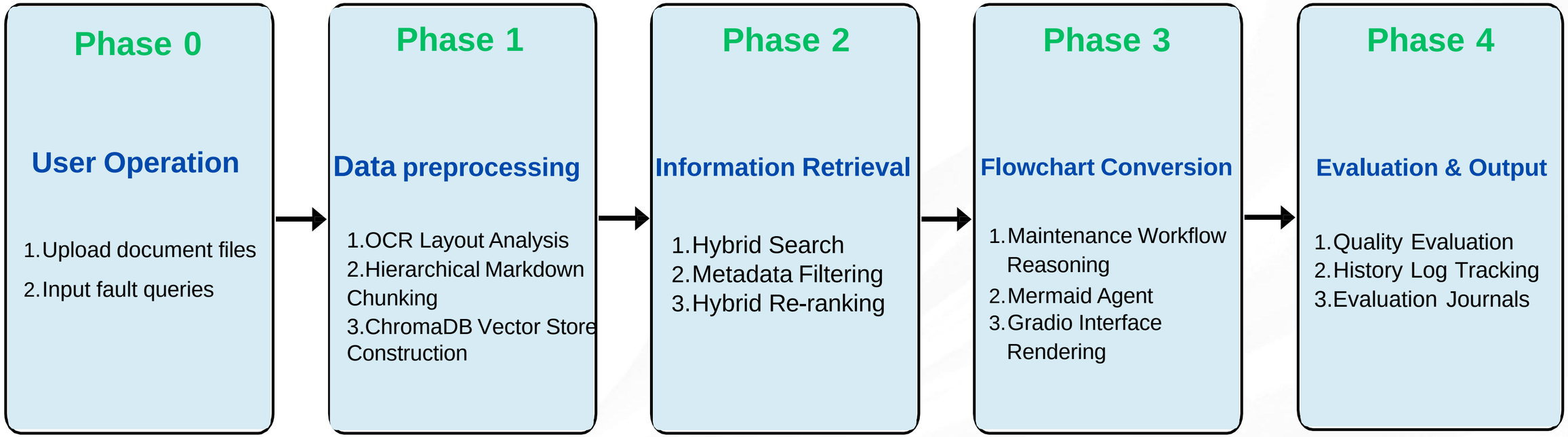
Major Pain Points in Traditional Factory O&M：

- **Scattered & Multi-modal Information**
- **Model Confusion & Hallucinations**
- **Tedious Maintenance Procedures**

Research Objectives：

- **Optimize Output Efficiency**
- **Enhance Output Accuracy**

System Architecture Design



Data Preprocessing

Comparison of contents	IBM Docling	EasyOCR	PaddleOCR
Table extraction capability	Execllent (TableFormer model, accuracy 97%)	❌ Poor (Can only output text blocks)	Good
Layout/Hierarchy Understanding	Execllent (Precise Markdown hierarchy)	❌ None	Good
Output format	Markdown, JSON, HTML, DocTags	Coordinates and text list	Coordinates and text list
Deployment difficulty	Simple (Simply install docling with pip)	Simple (Automatically download PyTorch models)	Medium (Highly dependent on certain packages in the environment)

```
{
    "ai_equipment": "萃取的設備或組件名稱 (如：逆變器)",
    "ai_error_code": "警報編號或故障碼 (如：E-045，若無則填'無')",
    "ai_category": "文件分類 (如：故障排除、安全警告、操作步驟)"
}
```

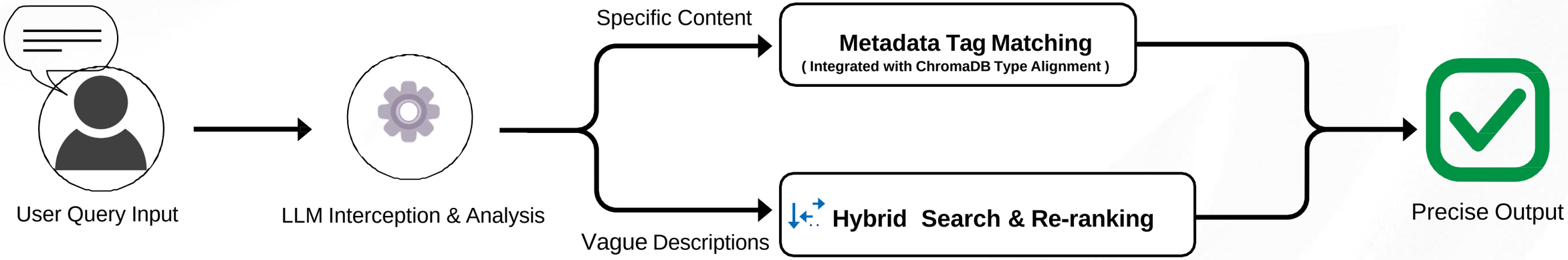
OCR Data Scan:
Scans complex tables to achieve complete data extraction.

Multi-level title semantic segmentation:
Based on the H1-H3 hierarchy, the complete context and maintenance procedures are preserved.

Structured Metadata Tags:
Before storing the file in the database, the file name and page/line number are encapsulated into metadata to enable front-end content comparison.

LLM (Automated Semantic Tagging)
Before data is imported into the database ➔ LLM annotation ➔ Force precise JSON output

Information Retrieval



Hybrid Search & Re-ranking

- **Hybrid Search**

  Executes keyword and vector (semantic) searches simultaneously to ensure both exact matches and vague queries are captured.

- **Hybrid Re-ranking**

  Utilizes a secondary ranking layer to resolve the critical RAG pain point of confusing highly similar equipment models (e.g., A-100 vs. A-101).

Mermaid Agent

Dynamic Visual Assistance: Dynamically generates Mermaid flowcharts from textual SOPs.

- JSON Array Extraction :
  1. Reconstructs RAG-retrieved text into a JSON format containing conditional branching arrays.
  2. Autonomously generates 1 to N troubleshooting branches based on manual complexity, fully adapting logic dynamically to the text.

- Defensive Character Processing :
  1. Filters and replaces dangerous characters (such as { },[ ],( ), newlines, and quotes) that could cause Mermaid rendering failures.

Evaluation Metrics

Performs real-time Pass / Fail safety compliance reviews for every generation.

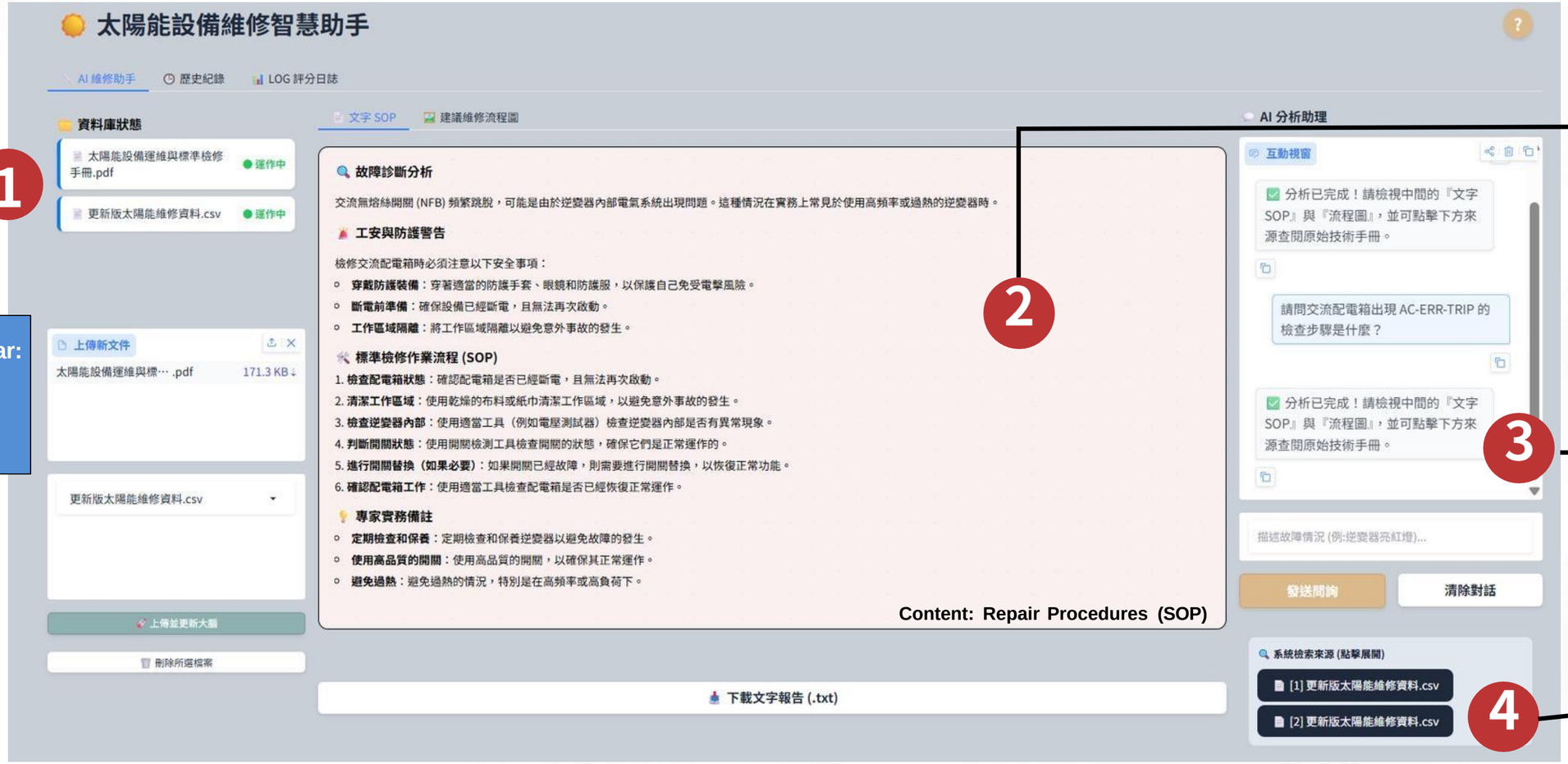
Evaluation Dimension	Judgment Criteria	Optimization Goal
Groundedness	Audits if the generated answer is completely traceable to the retrieved context.	**Eliminate Hallucinations**
Relevance	Evaluates if the system precisely addresses the user's inquiry.	**Reduce Redundant Info**
Safety	Checks for the inclusion of high-voltage alerts, protective gear reminders, etc.	**Comply with HSE Standards**

If any 'Fail' appears in the evaluation results, the following will be output.

⚠️查無相關手冊資訊，為確保設備安全，請勿進行未授權的拆卸或參數調整。

參考資料 https://learn.microsoft.com/zh-tw/azure/databricks/mlflow3/genai/eval-monitor/third-party-scorers/ragas

System UI Presentation



Database Status Bar:

Real-time monitoring of PDF/CSV vector store operations

Dynamic Textual SOP

Structured output of fault diagnosis and maintenance recommendations

Interactive Chat Window

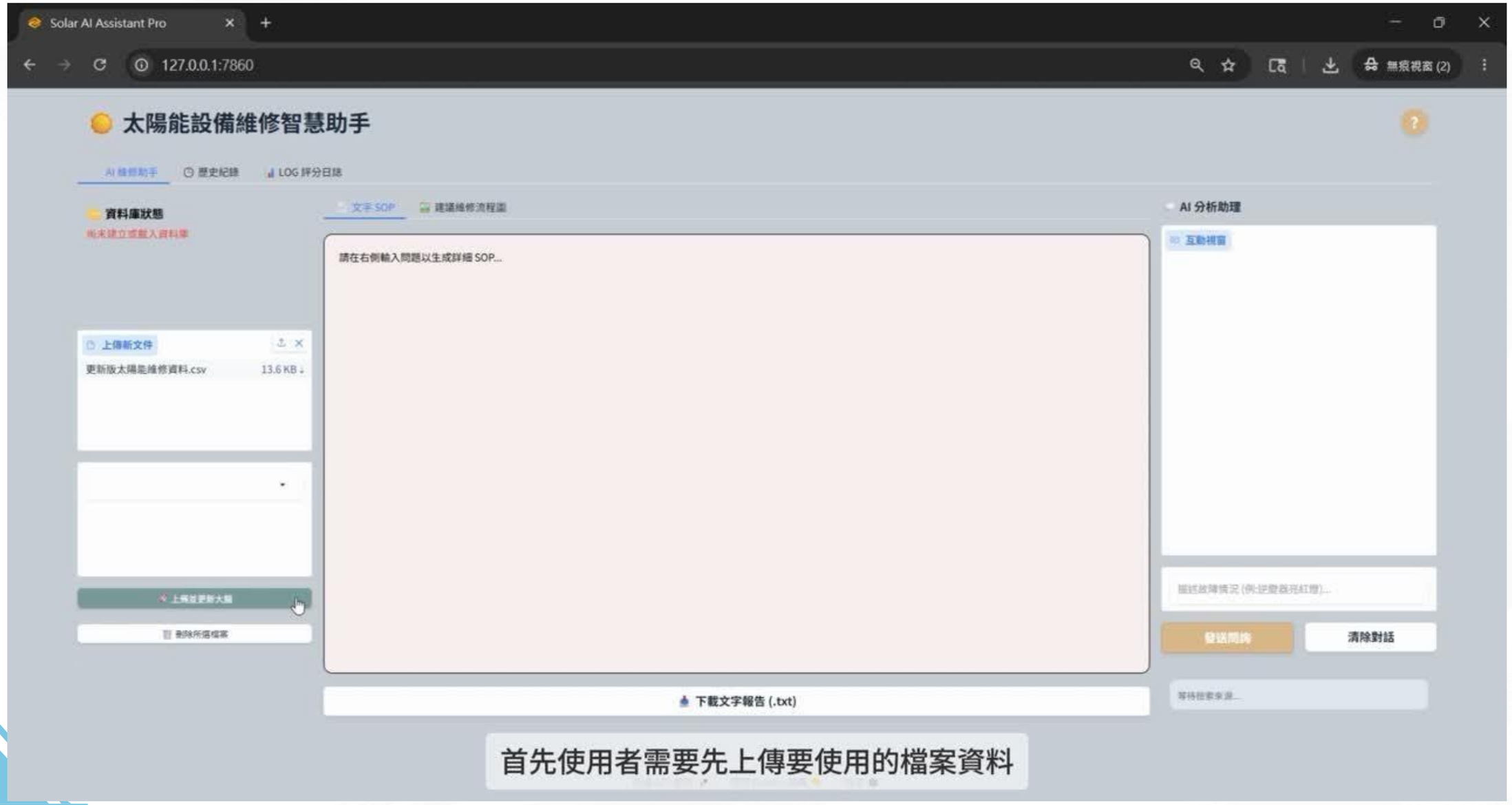
Supports continuous follow-up questions and real-time status updates.

Reference Source Viewer

Displays the original page of the technical manual upon clicking a source link

Content: Repair Procedures (SOP)

System Demonstration



『影片連結』

Conclusion & Future Outlook

Core Achievements :

ChromaDB Fail-safe Querying: Automatically aligns data types between filters and metadata during runtime, completely resolving missed retrievals caused by format mismatches.

System Efficiency Boost: Successfully reduced Operations & Maintenance (O&M) time by **55.6%**

Future Outlook

Multi-modal Visual Understanding: Integrate vision-language models (VLMs) to allow field technicians to take photos of equipment faults on-site.

Automated Retrieval: Automatically parse visual features and directly trigger the core RAG reasoning engine developed in this project, eliminating the tedious process of manual text input in the field.

Task assignment table

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Coding / Programming	Architecture Discussion	Data Collection
Poster & Video Production	Video Assistance	Presentation Preparation
Presentation Preparation	Presentation Preparation	System Testing

THANK YOU